

Linear Algebra Bootcamp 4

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Problem 1. Let $A : V \rightarrow W$ be a linear map between finite dimensional F vector spaces. For a subspace S in W show that $\dim A^*(S)$ is less than or equal to $\dim(S) + \dim(\ker A)$, with equality iff S is contained in the image of V .

Claim 1a. $\dim A^*(S) \leq \dim S + \dim \ker A$

Proof. By Linear Algebra Bootcamp 3 problem 5, since S is a subspace in W , we know $\ker A \subset A^*(S)$.

By Linear Algebra Bootcamp 2 problem 2, we know $\dim V = \dim A_*(V) + \dim \ker A$.

Let $B : A^*(S) \rightarrow W$ be defined as A restricted on the domain of $A^*(S)$. By Linear Algebra Bootcamp 2 problem 2, we $\dim A^*(S) = \dim B_*(A^*(S)) + \dim \ker B$. Since $\ker A \subset A^*(S)$, and B is defined as A with its domain restricted, $\dim \ker B = \dim \ker A$. So $\dim A^*(S) = \dim B_*(A^*(S)) + \dim \ker A$. Since we know $A^*(S)$ is a subspace of V with the kernel, by Linear Algebra Bootcamp 3 problem 5, we know $B_*(A^*(S))$ is a subspace.

For all $s \in B_*(A^*(S))$, by definition of image, there exists $v \in A^*(S)$ s.t $Bv = Av = s$. By definition of a preimage, $Av = s \in S$. Thus for all $s \in B_*(A^*(S))$, $s \in S$. Thus $B_*(A^*(S)) \subset S$.

1. If $S \subset A_*(V)$, for all $s \in S$ there exists a $v \in V$ s.t $Av = s$. Thus by definition of preimage, $v \in A^*(S)$. By the definition of a image, $Bv = Av = s \in B_*(A^*(S))$. Thus for all $s \in S$, $s \in B_*(A^*(S))$. Thus $S \subset B_*(A^*(S))$. Since $B_*(A^*(S)) \subset S$ and $S \subset B_*(A^*(S))$, then $S = B_*(A^*(S))$. Then $\dim S = \dim B_*(A^*(S))$. Thus $\dim A^*(S) = \dim S + \dim \ker A$.
2. If $S \not\subset A_*(V)$, there exists a $s \in S$ s.t there exists no $v \in V$ s.t $Av = s$. Since there exists no $v \in V$ and since $A^*(S) \subset V$ by definition, thus there exists no $v \in A^*(S)$ s.t $Bv = Av = s$. Thus by definition of image, $s \notin B_*(A^*(S))$. Thus $S \not\subset B_*(A^*(S))$. Thus since $B_*(A^*(S)) \subset S$ and $S \not\subset B_*(A^*(S))$, by Linear Algebra Bootcamp 1 problem 7, $\dim B_*(A^*(S)) < \dim S$. Thus $\dim A^*(S) < \dim S + \dim \ker A$.

Thus $\dim A^*(S) \leq \dim S + \dim \ker A$. □

Claim 1b. $\dim A^*(S) = \dim S + \dim \ker A \iff S \subset A_*(V)$

Proof. ($\implies$): Suppose $\dim A^*(S) = \dim S + \dim \ker A$. By Linear Algebra Bootcamp 2 problem 2, we know $\dim A^*(S) = \dim B_*(A^*(S)) + \dim \ker A$. Thus $\dim B_*(A^*(S)) = \dim S$. Assume the contrary, $S \not\subset A_*(V)$, thus there exists a $s \in S$ s.t there exists no $v \in V$ s.t $Av = s$. Since there exists no $v \in V$ and since $A^*(S) \subset V$ by definition, thus there exists no $v \in A^*(S)$ s.t $Bv = Av = s$. Thus by definition of image, $s \notin B_*(A^*(S))$. Thus $S \not\subset B_*(A^*(S))$. Thus since $B_*(A^*(S)) \subset S$ and $S \not\subset B_*(A^*(S))$, by Linear Algebra Bootcamp 1 problem 7, $\dim B_*(A^*(S)) < \dim S$. This forms a contradiction, thus $S \subset A_*(V)$.

($\impliedby$): Proven above. Refer above. □

Problem 2. Let $A : V \rightarrow W$ and $B : W \rightarrow Z$ be linear maps. Show $\dim \ker(BA) \leq \dim \ker(A) + \dim \ker(B)$ with equality iff $\ker B$ is contained in $A_*(V)$.

Claim 2a. $\dim \ker(BA) \leq \dim \ker(A) + \dim \ker(B)$

Proof. By Linear Bootcamp 2 problem 2, we know $\dim V = \dim(BA_*(V)) + \dim \ker BA$, $\dim V = \dim A_*(V) + \dim \ker A$, and $\dim W = \dim B_*(W) + \dim \ker B$. Thus we know $\dim \ker BA = \dim V - \dim(BA_*(V))$, $\dim \ker A = \dim V - \dim A_*(V)$, and $\dim \ker B = \dim W - \dim B_*(W)$.

Thus $\dim \ker A + \dim \ker B - \dim \ker BA = \dim V - \dim A_*(V) + \dim \ker B - (\dim V - \dim(BA_*(V))) = \dim \ker B - \dim A_*(V) + \dim(BA_*(V))$. By definition of BA and image, $BA_*(V) = \{BAv|v \in V\}$ and $B_*(A_*(V)) = B_*(\{Av|v \in V\}) = \{Bw|w \in \{Av|v \in V\}\} = \{B(Av)|v \in V\} = \{BAv|v \in V\}$, thus $BA_*(V) = B_*(A_*(V))$. Restrict the domain of B to $A_*(V)$ and call this function B' , by Linear Bootcamp 2 problem 2 we know $\dim A_*(V) = \dim B'_*(A_*(V)) + \dim \ker B'$. $B'_*(A_*(V)) = B_*(A_*(V))$ by definition. By definition of B' , $\ker B'$ is the kernel of B inside the domain $A_*(V)$. Thus $\ker B' = \ker B \cap A_*(V)$. Thus $B_*(A_*(V)) = \dim A_*(V) - \dim(\ker B \cap A_*(V))$. Thus $\dim \ker B - \dim A_*(V) + \dim(BA_*(V)) = \dim \ker B - \dim A_*(V) + \dim A_*(V) - \dim(\ker B \cap A_*(V)) = \dim \ker B - \dim(\ker B \cap A_*(V))$. By definition of intersection, $\ker B \cap A_*(V)$ is the set of elements which are in both $\ker B$ and $A_*(V)$, thus $\ker B \cap A_*(V) \subseteq \ker B$. By Linear Algebra Bootcamp 1 problem 7, $\dim \ker B \cap A_*(V) \leq \dim \ker B$. Thus $\dim \ker A + \dim \ker B - \dim \ker BA = \dim \ker B - \dim \ker B \cap A_*(V) \geq 0$. Thus $\dim \ker A + \dim \ker B - \dim \ker BA \geq 0$ and $\dim \ker BA \leq \dim \ker A + \dim \ker B$. $\square$

Claim 2b. $\dim \ker(BA) = \dim \ker(A) + \dim \ker(B) \iff \ker B \subseteq A_*(V)$

Proof. The proof is very similar to the one above.

($\implies$): Suppose $\dim \ker(BA) = \dim \ker(A) + \dim \ker(B)$. Then $\dim \ker(A) + \dim \ker(B) - \dim \ker(BA) = 0$.

By Linear Bootcamp 2 problem 2, we know $\dim V = \dim(BA_*(V)) + \dim \ker BA$, $\dim V = \dim A_*(V) + \dim \ker A$, and $\dim W = \dim B_*(W) + \dim \ker B$. Thus we know $\dim \ker BA = \dim V - \dim(BA_*(V))$, $\dim \ker A = \dim V - \dim A_*(V)$, and $\dim \ker B = \dim W - \dim B_*(W)$.

Thus $\dim \ker A + \dim \ker B - \dim \ker BA = \dim V - \dim A_*(V) + \dim \ker B - (\dim V - \dim(BA_*(V))) = \dim \ker B - \dim A_*(V) + \dim(BA_*(V))$. By definition of BA and image, $BA_*(V) = \{BAv|v \in V\}$ and $B_*(A_*(V)) = B_*(\{Av|v \in V\}) = \{Bw|w \in \{Av|v \in V\}\} = \{B(Av)|v \in V\} = \{BAv|v \in V\}$, thus $BA_*(V) = B_*(A_*(V))$. Restrict the domain of B to $A_*(V)$ and call this function B' , by Linear Bootcamp 2 problem 2 we know $\dim A_*(V) = \dim B'_*(A_*(V)) + \dim \ker B'$. $B'_*(A_*(V)) = B_*(A_*(V))$ by definition. By definition of B' , $\ker B'$ is the kernel of B inside the domain $A_*(V)$. Thus $\ker B' = \ker B \cap A_*(V)$. Thus $B_*(A_*(V)) = \dim A_*(V) - \dim(\ker B \cap A_*(V))$. Thus $\dim \ker B - \dim A_*(V) + \dim(BA_*(V)) = \dim \ker B - \dim A_*(V) + \dim A_*(V) - \dim(\ker B \cap A_*(V)) = \dim \ker B - \dim(\ker B \cap A_*(V))$. By definition of intersection, $\ker B \cap A_*(V)$ is the set of elements which are in both $\ker B$ and $A_*(V)$, thus $\ker B \cap A_*(V) \subseteq \ker B$. By Linear Algebra Bootcamp 1 problem 7, $\dim \ker B \cap A_*(V) \leq \dim \ker B$. Thus $\dim \ker B - \dim \ker B \cap A_*(V) = 0$. Thus $\dim \ker B = \dim \ker B \cap A_*(V)$. Since $\ker B \cap A_*(V) \subseteq \ker B$, by definition of intersection this is only possible if $\ker B \cap A_*(V) = \ker B$, so all of $\ker B$ is in $A_*(V)$. Thus $\ker B \subseteq A_*(V)$.

($\impliedby$): Suppose $\ker B \subseteq A_*(V)$. Then by the above proof, $\dim \ker A + \dim \ker B - \dim \ker BA = \dim \ker B - \dim \ker B \cap A_*(V) \geq 0$. If $\ker B \subseteq A_*(V)$, then $\ker B \cap A_*(V) = \ker B$ by definition of intersection. Then $\dim \ker B \cap A_*(V) = \dim \ker B$. Thus $\dim \ker A + \dim \ker B - \dim \ker BA = \dim \ker B - \dim \ker B \cap A_*(V) = 0$. $\square$

Problem 3. Show that every injective (resp surjective) linear map has a linear left (resp right) inverse.

Claim 3a. Every injective linear map has a linear left inverse.

Proof. Let V, W be F -vector spaces. Let $f : V \rightarrow W$ be an injective linear map. Since $f_*(V) \cong V/\ker f$ and $V/\ker f$ is a vectorspace, $f_*(V)$ must also be a vectorspace and since $f_*(V) \subseteq W$ it is a subspace of W . Let the basis of $f_*(V)$ be the set $B_{f_*(V)} = \{b_1, \dots, b_n\}$. We can expand this basis to get the basis of W , $B_W = \{b_1, \dots, b_n, w_1, \dots, w_m\}$.

Define $g : W \rightarrow V$, as the following:

For all $w \in W$ such that $w \in f_*(V)$. By definition of image, we know there exists a $v \in V$ s.t. $f(v) = w$. Thus let $g(w) = v$. For all $w \in W/f_*(V)$, by definition of basis, B_W spans W . Thus $w = \sum_{i=1}^n a_i b_i + \sum_{j=1}^m c_j w_j$ where $a_i, b_j \in F$. Since $\sum_{i=1}^n a_i b_i$ is a linear combination of the basis elements of $f_*(V)$, by definition of a subspace $\sum_{i=1}^n a_i b_i = x \in f_*(V)$. Thus $g(w) = g(\sum_{i=1}^n a_i b_i) = g(x)$ where since $x \in f_*(V)$ we use the case above.

$\forall v \in V$, $(g \circ f)(v) = g(f(v))$. Since f is injective, if $f(a) = f(b) \implies a = b$. Thus $f^*(\{f(v)\}) = v$. So $g(f(v)) = v = id_V(v)$. So g is a left inverse.

$\forall v, w \in W$ and $p, q \in F$, $v = \sum_{i=1}^n a_i b_i + \sum_{j=1}^m c_j w_j$ and $w = \sum_{i=1}^n d_i b_i + \sum_{j=1}^m e_j w_j$. $g(pv + qw) = g(p \sum_{i=1}^n a_i b_i + q \sum_{i=1}^n d_i b_i) = g(\sum_{i=1}^n (pa_i + qd_i) b_i)$. Let $x = \sum_{i=1}^n (pa_i + qd_i) b_i$, $x \in f_*(V)$ thus there exists a $u \in V$ such that $f(u) = x$. $g(\sum_{i=1}^n (pa_i + qd_i) b_i) = g(x) = u = \sum_{i=1}^n (pa_i + qd_i) v_i$, because f is linear. $pg(v) + qg(w) = pg(\sum_{i=1}^n a_i b_i) + qg(\sum_{i=1}^n d_i b_i)$. Let $y = \sum_{i=1}^n a_i b_i$ and $z = \sum_{i=1}^n d_i b_i$. $pg(v) + qg(w) = pg(y) + qg(z)$. There exists a $r, s \in V$, such that $f(r) = y$ and $f(s) = z$. Since f is linear, $r = \sum_{i=1}^n a_i v_i$ and $s = \sum_{i=1}^n d_i v_i$. Thus $pg(y) + qg(z) = pr + qs = \sum_{i=1}^n (pa_i + qd_i) v_i = u$. Thus g is linear $\square$

Claim 3b. Every surjective linear map has a linear right inverse.

Proof. Let V, W be F -vector spaces. Let $f : V \rightarrow W$ be a surjective linear map.

Since f is surjective $f_*(V) = W$. Then the preimage of every element in W is nonempty. Let $g : W \rightarrow V$ such that for all $w \in W$, $g(w)$ randomly chooses an element in $f^*(\{w\})$.

$\forall v, w \in V$, since f is surjective there exists $a, b \in V$ such that $f(a) = v$ and $f(b) = w$. Furthermore since f is linear, $f(a + b) = f(a) + f(b) = v + w$. By definition $f(g(v + w)) = v + w$. $v + w = f(a + b)$. $g(v + w) = a + b = g(v) + g(w)$. For all $c \in F$, since f is linear $cf(a) = cv = f(ca)$. Thus $g(cv) = ca = cg(v)$. Thus g is linear. $\square$

Problem 4. Let K be a subspace of an F -vector space W . Show that K is the kernel of some linear map (from W to some F -vector space). Let K be a subspace of F^m . Show that there is an $n \times m$ matrix with kernel K . Show further there is an $m \times m$ matrix with kernel K . Explain why every subspace of F^m is the solution set to an $m \times m$ system of homogeneous linear equations.

Claim 4a. Let K be a subspace of an F -vector space W . Show that K is the kernel of some linear map.

Proof. Let V be an arbitrary F -vector space where $\dim V \geq \dim W - \dim K$. This choice for the dimension, ensures that there is enough "room" in V to map the entirety of W while leaving enough dimensions to capture the kernel of the map. By definition of a vector space, there must exist a $0_v \in V$ which is the additive identity.

Let set $B_K = \{k_1, \dots, k_m\}$ be the basis set of subspace $K \subseteq W$. By definition of a basis, B_K is linearly independent and spans K . Moreover by definition of a subspace, $B_K \subset W$. By Linear Algebra Bootcamp 2 problem 1, $B_K \subseteq B_W$ where B_W is a basis of W . Let $B_W = \{w_1, \dots, w_n, k_1, \dots, k_m\}$. Let $B_V = \{v_1, \dots, v_p\}$ be the basis set of V where by definition of dimension, $p \geq \dim W - \dim K$. Define a linear map $g : W \rightarrow V$ where $g(w_i) = v_j$ for all $i \in [1, \dots, n]$ and some $j \in [0, \dots, p]$ and $g(k_i) = 0$ for all $i \in [1, \dots, m]$. Thus all the basis vectors of K get mapped to 0. Since B_K spans K by definition, $\forall k \in K$, $g(k) = g(\sum_{i=1}^m a_i k_i) = \sum_{i=1}^m a_i g(k_i) = 0$ for all $a_i \in F$. Thus $K \subseteq \ker g$. However since the

basis vectors in B_W which map to 0, are the basis vectors of K , then $\ker g \subseteq K$. Thus $K = \ker g$. Thus K is the kernel of some linear map with domain of W . $\square$

Claim 4b. *Let K be a subspace of F^m . Show that there is an $n \times m$ matrix with kernel K .*

Proof. Let $W = F^m$ and let K be a subspace of W . Let $V = F^n$ where $\dim V = n \geq \dim W - \dim K = m - \dim K$. By the previous proof, we know there exists a $f : W \rightarrow V$ where kernel of f is K . Since linear maps between vector spaces can also be written as a matrix A . By definition of A , $\ker f = \ker A = K$. Furthermore, by definition of the induced matrix the size of A is $\dim V \times \dim W = n \times m$ thus $A \in M_{n \times m}$. Thus there exists a $A \in M_{n \times m}$ such that $\ker A = K$. $\square$

Claim 4c. *Let K be a subspace of F^m . Show that there is an $m \times m$ matrix with kernel K .*

Proof. Let $W = F^m$ and K be a subspace of W . Let $V = F^n$ where $\dim V = n \geq \dim W - \dim K = m - \dim K$. If $n = m$, then this condition is satisfied because $m \geq m - \dim K$. Thus by the previous proof there exists a matrix $A \in M_{n \times m}$, and thus a matrix $A \in M_{m \times m}$ s.t $\ker A = K$. $\square$

Claim 4d. *Explain why every subspace of F^m is the solution set to an $m \times m$ system of homogeneous linear equations.*

Solution. K in the previous proof can be any subspace of F^m . Thus you can create a matrix $A \in M_{m \times m}$ such that its kernel is K for any K . Thus all $x \in K$ satisfy $Ax = 0$. And any $x \in F^m$ that satisfies $Ax = 0$ is in the kernel of A and thus in K . $\blacksquare$

Problem 5. *Let A be an $n \times m$ matrix of rank k . Let $r_1, \dots, r_k$ be rows of A spanning the row space, and $c_1, \dots, c_k$, columns of A spanning the column space. Let S be the $k \times k$ matrix you get using these k rows and k columns (the k^2 entries are the intersections of the k rows with the k columns). Show S is invertible. Hint. Let A' be the $k \times m$ matrix you get by dropping all the rows other than $r_1, \dots, r_k$. Note each column, c , of A naturally gives a column, c' , of A' . Explain why A' has rank k , and $c'_1, \dots, c'_k$ span the column space of A' . Explain why we can replace A by A' , and so can assume from the start that the rows of A are independent. Now do the analogous thing with column to reduce to the case when A is square with independent rows and columns.*

Proof. Consider the matrix $A' \in M_{k \times m}$ obtained by keeping only rows $r_1, \dots, r_k$ of A . Each column, c , of A naturally gives a column, c' , of A' . Since each column c of A naturally gives a column c' . Each column $c_1, \dots, c_k$ has a natural analog in A' , $c'_1, \dots, c'_k$. Furthermore since $c_1, \dots, c_k$ spans column space of A , $c'_1, \dots, c'_k$ spans the column space of A' . By definition of rank, A' has rank k . We can replace A' with A , because the residual rows of A are linear combinations of the rows in $r_1, \dots, r_k$. Thus those rows add no new information. Now do the analogous thing for A' so the square and since its rows and columns are basis elements in the column space and row space of A they are linearly independent. Since the rank of A is equal to the number of rows and columns of A and A is square, by definition of invertible matrices, A is an invertible matrix. $\square$

Problem 6. *Explain the statement: Up to choosing bases the only linear maps between finite dimensional vector spaces are the maps $A : F^m \rightarrow F^n$ given by $A(x_1, \dots, x_n) = (x_1, \dots, x_k, 0, \dots, 0)$ for some k .*

Solution. By Linear Algebra Bootcamp 1 problem 7, there exists isomorphisms $R : F^n \rightarrow F^n$ and $C : F^m \rightarrow F^m$ such that RAC is a "diagonal matrix" which is a matrix with a identity matrix of some size flanked by 0s. R and C represent change of basis. Thus for all $e_i^m \in F^m$ which is the standard basis, $RACe_i^m = e_i^n$ or $RACe_i^m = 0$. Thus for all $v \in F^m$, $RACv = RAC(\sum_{i=1}^m a_i e_i^m) = \sum_{i=1}^m a_i (RACe_i^m)$, is a vector with a_i in the i th position if $RACe_i^m \neq 0$. Thus the only choice afforded to us is the choice of bases in the vector space and thus the only linear maps are the maps $A(x_1, \dots, x_n) = (x_1, \dots, x_k, 0, \dots, 0)$ for some k . ■